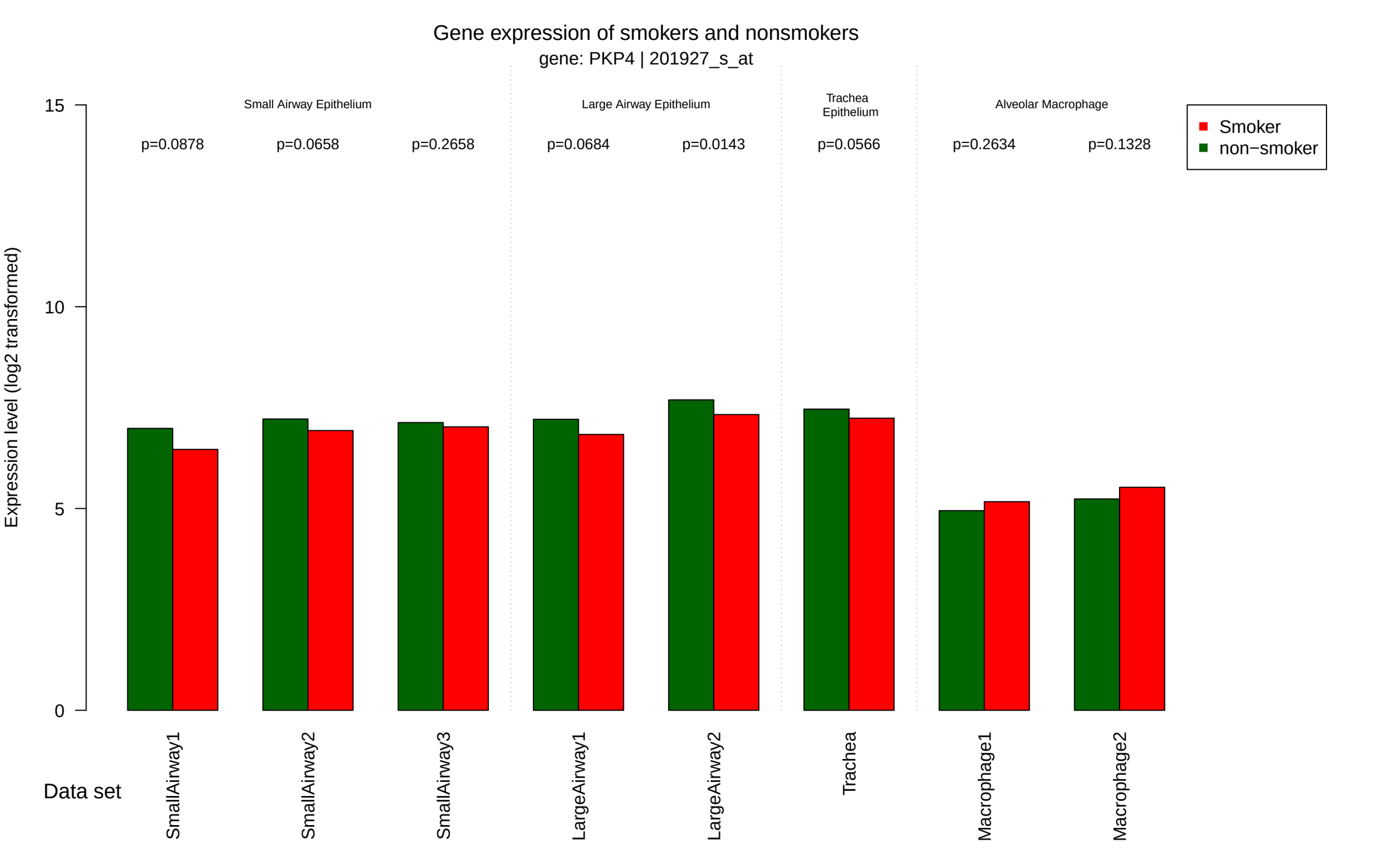
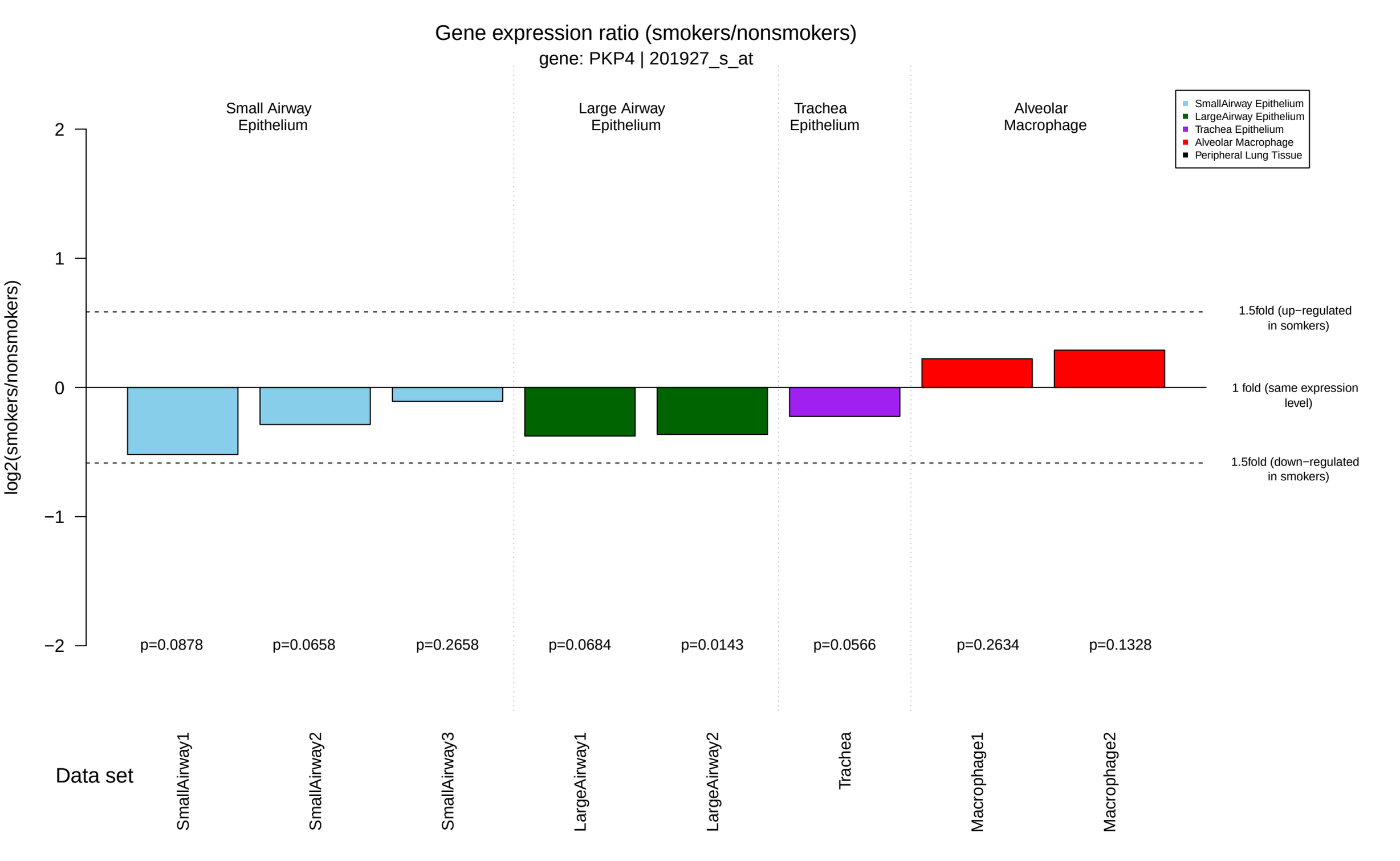
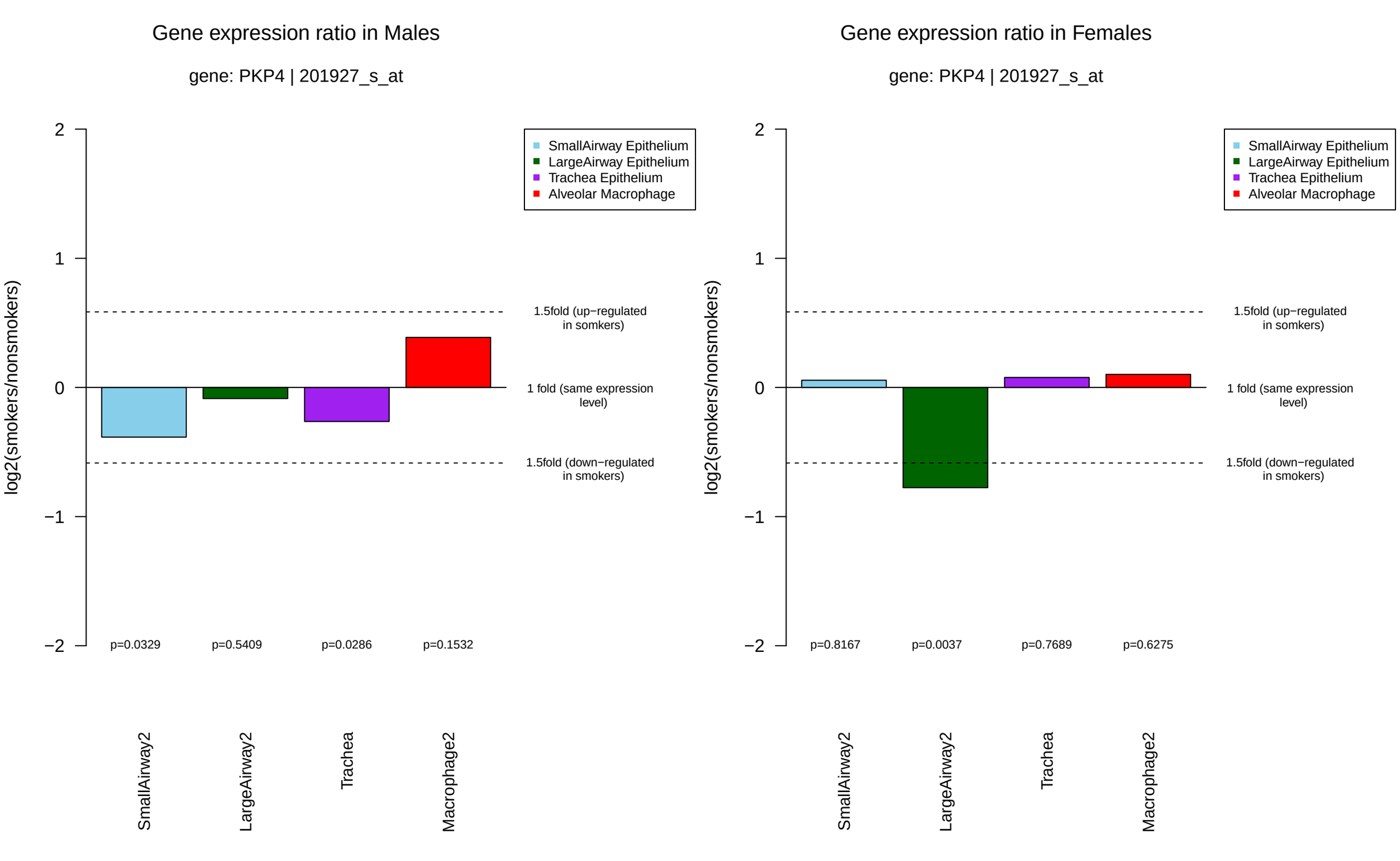
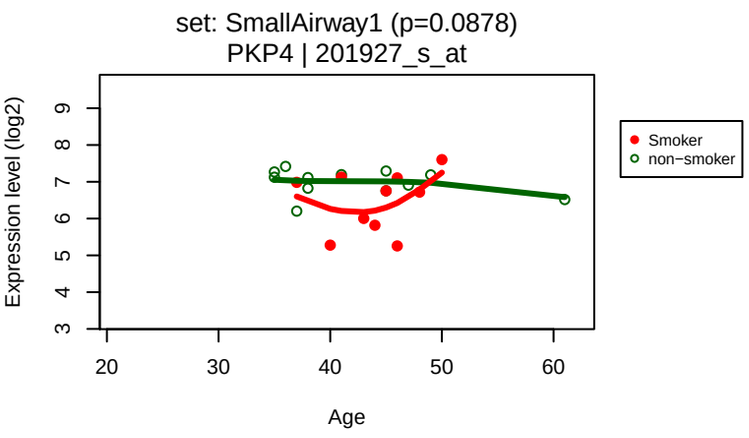
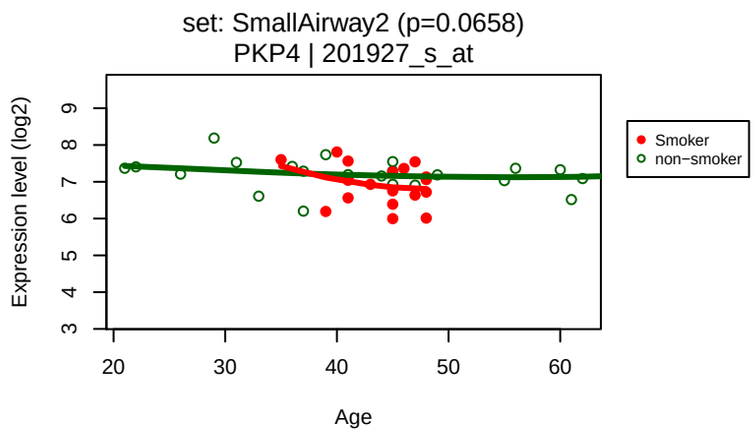
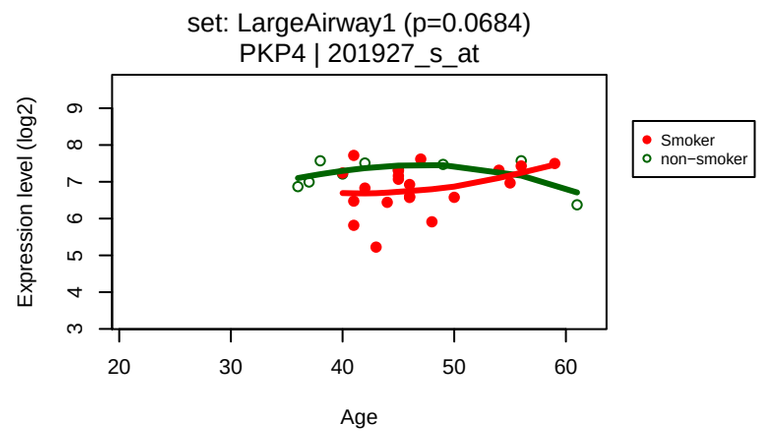

Gene expression of smokers and nonsmokers
gene: PKP4 | 201927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0878
p=0.0658
p=0.2658
p=0.0684
p=0.0143
p=0.0566
p=0.2634
p=0.1328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

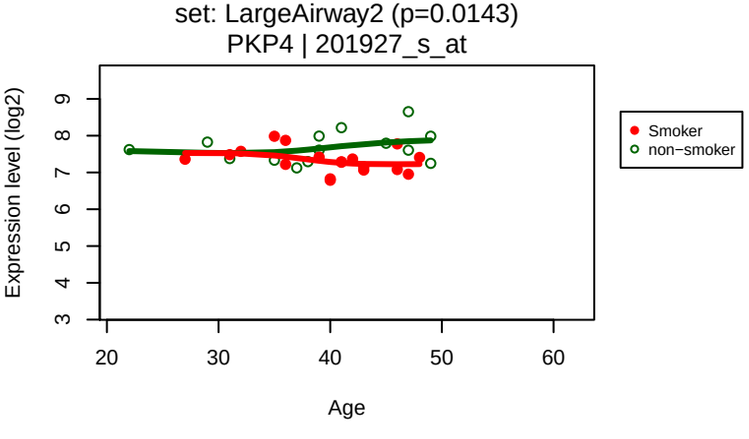

Gene expression ratio (smokers/nonsmokers)
gene: PKP4 | 201927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0878
p=0.0658
p=0.2658
p=0.0684
p=0.0143
p=0.0566
p=0.2634
p=0.1328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

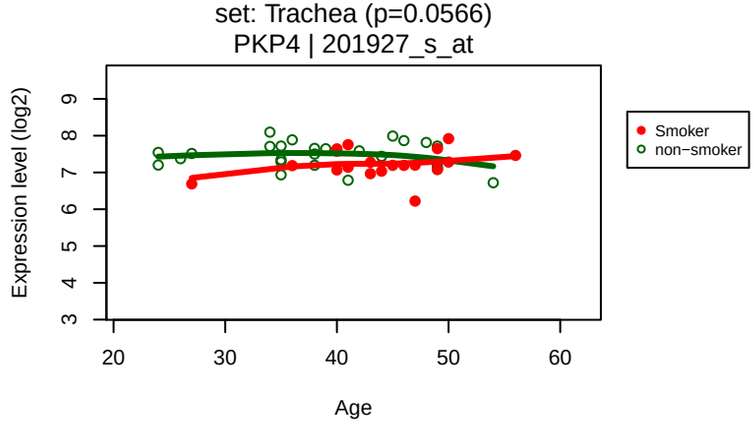

Gene expression ratio in Males
gene: PKP4 | 201927_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0329
p=0.5409
p=0.0286
p=0.1532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKP4 | 201927_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8167
p=0.0037
p=0.7689
p=0.6275
SmallAirway2
LargeAirway2
Trachea
Macrophage2

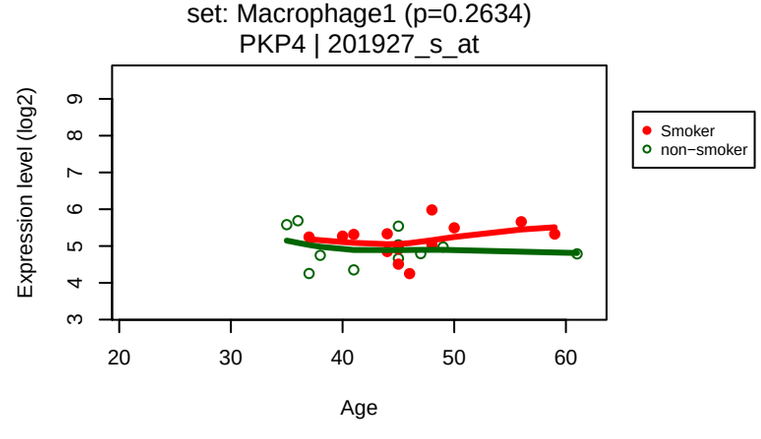

set: SmallAirway1 (p=0.0878)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker

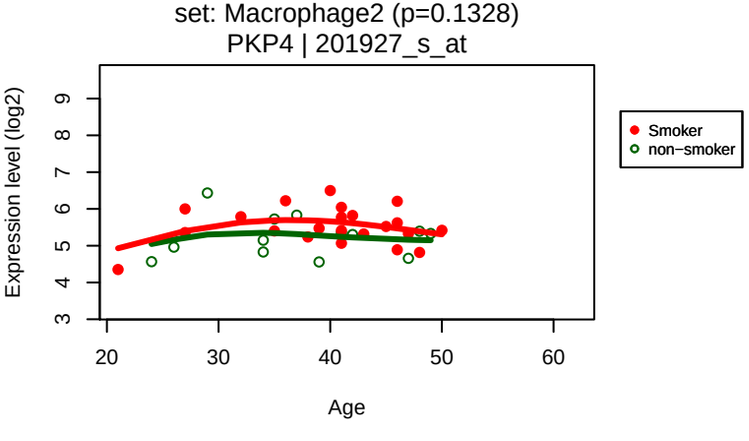

set: SmallAirway2 (p=0.0658)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker

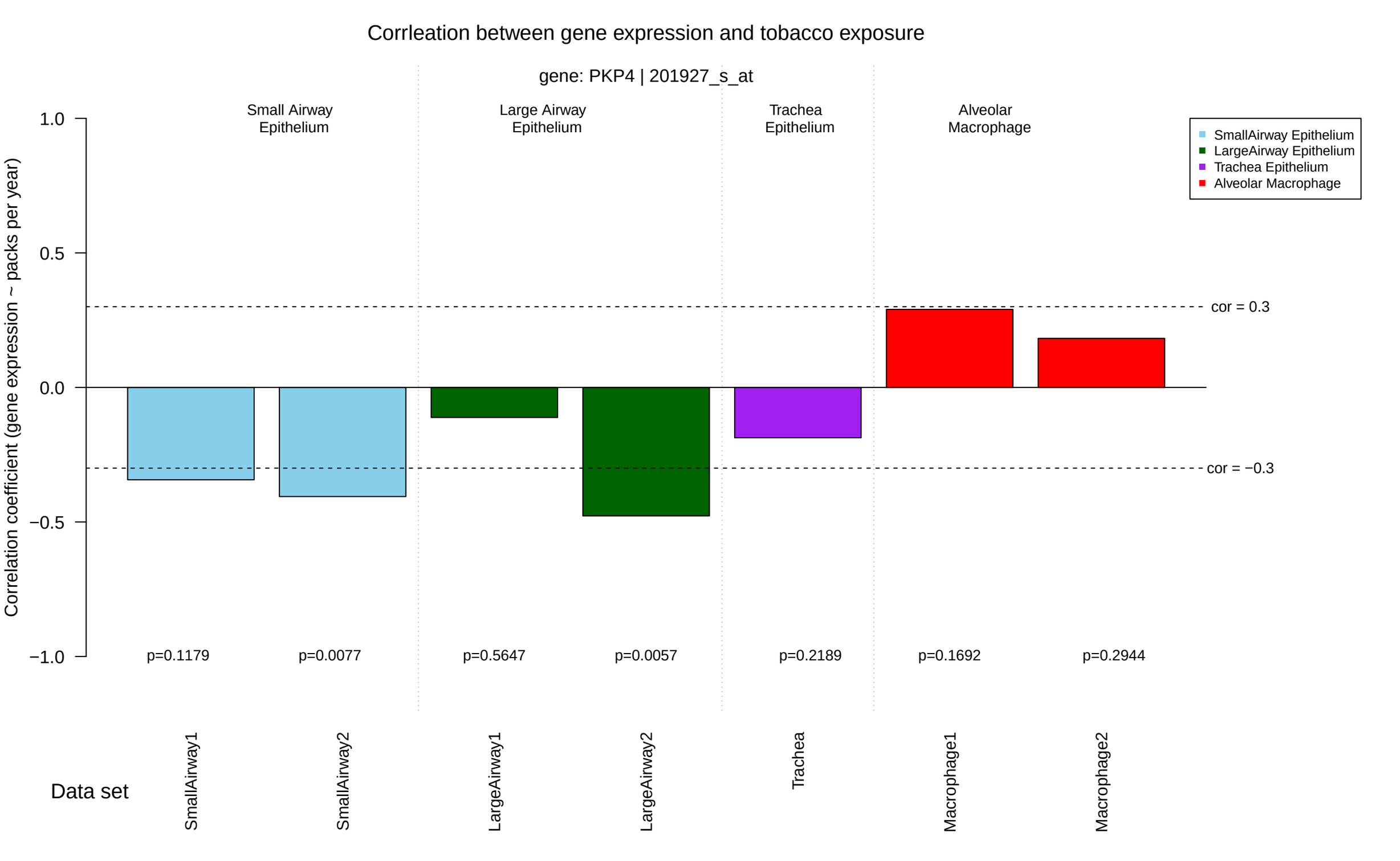

set: LargeAirway1 (p=0.0684)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0143)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0566)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2634)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1328)
PKP4 | 201927_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKP4 | 201927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1179
p=0.0077
p=0.5647
p=0.0057
p=0.2189
p=0.1692
p=0.2944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2